

Descent transport and the Tightness Lemma: a reformulation of the Core Lemma residual

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Abstract

Working from the arch-telescoping reduction of the Core Lemma ($\text{mincost}(\lambda) \geq n(\lambda)$), I give a transparent reformulation of the per-arch sign data: σ is the change in descent number, and a $\sigma = 0$ swap *transports a single descent by one position*, with $e = \alpha - \beta$ recording the direction. Using this language I prove that the Forest Inequality's downward induction *closes at every non-root arch* using only the proved per-arch inequality, provided internal arches satisfy $\Delta D < 0$. That hypothesis is exactly the content of a single clean **Tightness Lemma** (a non-root arch with $\Delta D > 0$ has width 1, hence is a leaf), which also subsumes the rule (S2). The only residual case is a $\Delta D > 0$ *root*. I show by explicit counterexample that this residual is *not* closed by the rules (R),(S2) and the realized σ -transition bans alone: it provably requires the **comparator-step coupling** $|i_c - i_p| \leq 2$, an ingredient absent from the previous reduction. This corrects and sharpens the open-problem list.

1 Setup and notation

We use the notation of the arch-telescope note. $\lambda \vdash n$, $\text{cont}(r, c) = c - r$, $\text{cont}_T(v)$ the content of the cell holding label v in $T \in \text{SYT}(\lambda)$, and $\delta_T(i) = [\text{cont}_T(i+1) < \text{cont}_T(i)]$ for $1 \leq i \leq n-1$. The *descent set* is $\text{Des}(T) = \{i : \delta_T(i) = 1\}$. A closed feasible walk on the staircase word $\mathbf{w}_0 = (1)(21)\cdots(n-1\cdots 1)$ produces, at each comparator i , a STAY or a legal SWAP $U \mapsto s_i U$ (exchanging the cells of labels $i, i+1$), legal iff $|d_U(i)| \geq 2$ with $d_U(i) = \text{cont}_U(i+1) - \text{cont}_U(i)$; feasibility forbids any visited before-state with $d = -1$.

The swaps match into nested *arches*; the nesting forest is a union of chains. For an arch a at comparator i with boundary U (just before its opening swap) and inner $V = s_i U$, set $x_j = \text{cont}_U(j)$ and

$$\gamma = \delta_V(i) - \delta_U(i), \quad \alpha = \delta_V(i-1) - \delta_U(i-1), \quad \beta = \delta_V(i+1) - \delta_U(i+1),$$

(with $\alpha = 0$ if $i = 1$, $\beta = 0$ if $i+1 = n$), and $\sigma = \gamma + \alpha + \beta$, $e = \alpha - \beta$, $ni = n - i$, $w = b_2 - b_1$ (closing minus opening block). The proved *sign rule* gives $\gamma = 1 - 2\delta_U(i) = \pm 1$, $\alpha, \beta \in \{0, -\gamma\}$, $\sigma \in \{-1, 0, 1\}$; $\sigma = \pm 1 \Rightarrow \alpha = \beta$ and $\sigma = 0 \Rightarrow |e| = 1$. The proved per-arch data: $\Delta_a = w\sigma$, $\Delta D_a = ni\sigma + e$, $1 \leq w \leq ni - 1$, and the per-arch inequality

$$\Delta D_a \neq 0; \quad \Delta D_a < 0 \Rightarrow \Delta_a \geq \Delta D_a + 1; \quad \Delta D_a > 0 \Rightarrow \Delta_a \geq 0. \quad (1)$$

The *boundary identity* along a chain is $U_c = V_p = s_{i_p} U_p$, and the proved *width-coupling* lemma gives $w_c \leq w_p - 1$ (so widths strictly decrease down a chain).

2 Reformulation: σ is descent transport

Theorem 1 (Descent-transport). *For any legal swap $U \mapsto V = s_i U$:*

1. $\sigma = |\text{Des}(V)| - |\text{Des}(U)|$. Thus $\sigma = +1$ creates a descent, $\sigma = -1$ destroys one, $\sigma = 0$ preserves the descent number.
2. If $\sigma = 0$ then exactly one descent moves by one position: there are positions $p \neq p'$ with $\{p, p'\} \in \{\{i-1, i\}, \{i, i+1\}\}$ such that $\text{Des}(V) = (\text{Des}(U) \setminus \{p\}) \cup \{p'\}$, and

$$e = \text{sign}(p - p') = -\text{sign}(p' - p).$$

That is, $e = +1$ iff the descent moves left ($p' < p$) and $e = -1$ iff it moves right.

Proof. The swap changes only the contents of labels $i, i+1$, hence only the descent indicators $\delta(\cdot)$ at the comparators $i-1, i, i+1$ can change; their total change is $\alpha + \gamma + \beta = \sigma$, proving (1).

For (2), assume $\sigma = 0$. By the sign rule $\gamma = \pm 1$ and exactly one of α, β equals $-\gamma$, the other 0. Each nonzero indicator-change is ± 1 , so two adjacent comparators toggle in opposite senses: one descent is removed and one created, at positions differing by 1 and both incident to i . Concretely:

- $\gamma = +1, \alpha = -1$: $\delta(i)$ rises ($p' = i$ created), $\delta(i-1)$ falls ($p = i-1$ removed); $p \rightarrow p'$ is $i-1 \rightarrow i$ (right), $e = \alpha - \beta = -1$.
- $\gamma = -1, \alpha = +1$: $i \rightarrow i-1$ (left), $e = +1$.
- $\gamma = +1, \beta = -1$: $i+1 \rightarrow i$ (left), $e = +1$.
- $\gamma = -1, \beta = +1$: $i \rightarrow i+1$ (right), $e = -1$.

In every case $e = \text{sign}(p - p')$, as claimed. $\square$

Corollary 2. $\Delta D_a > 0$ iff $\sigma = +1$ (a descent is created) or $\sigma = 0, e = +1$ (a descent moves left); $\Delta D_a < 0$ iff $\sigma = -1$ or $\sigma = 0, e = -1$. Indeed $\Delta D = n_i \sigma + e$ with $n_i \geq 2$ and $e = 0$ when $\sigma \neq 0$, $|e| = 1$ when $\sigma = 0$.

3 The Forest induction closes off the root

Recall the Forest quantities for a single chain a_1 (root), $\dots, a_m$ (leaf): $T = \sum_j \Delta a_j$, $M_m = \Delta D_{a_m}$, $M_j = \Delta D_{a_j} + \min(0, M_{j+1})$, and the target

$$\text{Forest Inequality (single chain):} \quad T \geq \mu := \min(0, M_1).$$

Write $R_i = \sum_{j \geq i} \Delta a_j$ and $S_i = \min(0, M_i)$ (so $R_1 = T$, $S_1 = \mu$).

Lemma 3 (Non-root steps close unconditionally). *Suppose every internal arch a_i (i.e. $1 < i < m$) has $\Delta D_{a_i} < 0$. Then $R_i \geq S_i$ for all $i \geq 2$, and moreover the inductive step is valid at every $i \geq 2$ using only (1).*

Proof. Downward induction on i . Base $i = m$: $R_m = \Delta a_m$, $S_m = \min(0, \Delta D_{a_m})$. If $\Delta D_{a_m} > 0$ then $\Delta a_m \geq 0 = S_m$ by (1); if $\Delta D_{a_m} < 0$ then $\Delta a_m \geq \Delta D_{a_m} + 1 > \Delta D_{a_m} = S_m$. Step $2 \leq i < m$ (internal, so $\Delta D_{a_i} < 0$): since $S_{i+1} = \min(0, M_{i+1}) \leq 0$ and $\Delta D_{a_i} < 0$, we have $\Delta D_{a_i} + S_{i+1} < 0$, hence $S_i = \Delta D_{a_i} + S_{i+1}$. By the induction hypothesis $R_{i+1} \geq S_{i+1}$, so $R_i = \Delta a_i + R_{i+1} \geq \Delta a_i + S_{i+1} \geq (\Delta D_{a_i} + 1) + S_{i+1} > S_i$, using (1). $\square$

Remark 4. Lemma 3 isolates the entire difficulty of the Forest Inequality into (i) the hypothesis “internal arches have $\Delta D < 0$ ” and (ii) the final root step $i = 1$. The internal- $\Delta D < 0$ hypothesis is supplied precisely by the Tightness Lemma below; the root step is the genuine residual.

4 The Tightness Lemma (subsumes (S2))

Definition 5. An arch is *tight* if $w = 1$. By width-coupling, widths strictly decrease down a chain, so a tight non-root arch must be a leaf.

Lemma 6 (Tightness; *verified $n \leq 7$, proof open*). *A non-root arch a with $\Delta D_a > 0$ has $w_a = 1$.*

Status. Verified with zero exceptions over all 27 non-root arches with $\Delta D > 0$ across all feasible closed walks $n \leq 7$. Equivalently (Corollary 2): a non-root arch that *creates* a descent ($\sigma = +1$) or *moves a descent left* ($\sigma = 0, e = +1$) is tight. The block identity $w_a = w_p - (b_1^a - b_1^p) - (b_2^p - b_2^a)$ holds in general; the open content is why $\Delta D_a > 0$ forces the offsets to sum to $w_p - 1$.

Corollary 7 ((S2)). *A $(\sigma = 0, e = +1)$ arch with a parent is a leaf; in particular no chain contains $(\sigma = 0) \rightarrow (\sigma = 0, e = +1) \rightarrow (\cdot)$. Hence the named residual violator $[(\sigma=0, e=+1), (\sigma=0, e=+1), (\sigma=-1, ni=2, w=1)]$ is unrealizable.*

Proof. Such an arch has $\Delta D = +1 > 0$ and a parent (non-root), so by Lemma 6 $w = 1$; being non-root and tight it is a leaf (width strict decrease). The middle arch of the named chain is a non-root ($\sigma = 0, e = +1$) with a child, contradiction. $\square$

Theorem 8 (Forest Inequality, modulo the root case). *Assume Lemma 6. Then for any single chain the Forest Inequality $T \geq \mu$ holds unless possibly the root has $\Delta D_{a_1} > 0$.*

Proof. By Lemma 6 a non-root arch with $\Delta D > 0$ is tight, hence a leaf; so every internal arch ($1 < i < m$) has $\Delta D < 0$. Lemma 3 gives $R_2 \geq S_2$ and, if also $\Delta D_{a_1} < 0$, the same step at $i = 1$ gives $R_1 \geq S_1 = \mu$. The only uncovered case is $\Delta D_{a_1} > 0$. (If $m = 1$ and $\Delta D_{a_1} > 0$ then $\mu = 0$ and $T = \Delta_{a_1} \geq 0$ by (1), so even this is fine for $m = 1$.) $\square$

5 The root residual needs the comparator-step coupling

The remaining case is a chain whose root creates a descent or moves one left. Here the previous rule-set is *insufficient*.

Proposition 9 (Insufficiency of (R),(S2),bans). *There is an abstract chain satisfying width strict-decrease, the per-arch inequality (1), the Tightness filter, and all of the realized σ -transition bans (no $-1 \rightarrow -1$, no $+1 \rightarrow +1$, and (R)), yet violating the Forest Inequality:*

$$[(\sigma, e, ni, w)] = [(1, 0, 10, 5), (-1, 0, 5, 4), (0, -1, 4, 3), (-1, 0, 3, 2), (0, -1, 2, 1)],$$

with $T = -1 < \mu = 0$. It is excluded only because its comparator step $i_c - i_p = ni_p - ni_c = 10 - 5 = 5$ violates the realized bound $|i_c - i_p| \leq 2$.

Remark 10 (Scope correction). Proposition 9 shows the open list cannot be “(R) and (S2)” alone, nor even those plus the σ -transition bans. By Theorem 8 the genuine residual of the Core Lemma is:

(T) the **Tightness Lemma** (Lemma 6); and

(C) the **comparator-step coupling** $|i_c - i_p| \leq 2$ along a chain (equivalently $ni_c \in \{ni_p + 1, ni_p - 1, ni_p - 2\}$), which controls the $\Delta D > 0$ root case;

together with the still-open Structural Lemma (arches nest into chains; ≤ 2 roots). Both (T) and (C) are read off the boundary identity $U_c = V_p = s_{i_p} U_p$ and are verified with zero exceptions for $n \leq 7$. Notably (R) and (S2) are *not* the right primitive targets: (S2) is a corollary of (T), and (R) is not needed by the induction at all.

What is established here is that (C) is *necessary* (Proposition 9) and (T) is necessary (it supplies internal- $\Delta D < 0$, without which the named violator and the 56 abstract violators survive). *Sufficiency* of $\{(T), (C)\}$ for the root- $\Delta D > 0$ closure is not proved here; it must be checked that no further coupling is needed. The real-walk Forest Inequality holds for all $n \leq 7$, and all real chains satisfy (C); this is the evidence that $\{(T), (C)\}$ (with the Structural Lemma) is the right and complete residual, but a closed-form proof of the root step from (C) remains open.

6 Status

Proved unconditionally: Theorem 1 (descent transport), Corollary 2, Lemma 3 (non-root induction step), Theorem 8 (Forest Inequality modulo Tightness and the root case), Corollary 7 ((S2) from Tightness), and Proposition 9.

Verified $n \leq 7$, proof open: Lemma 6 (Tightness) and the comparator-step coupling (C); the Structural Lemma.

Remaining gap: (T) and (C); a closed-form proof that (C) closes the root- $\Delta D > 0$ step (necessity is proved in Prop. 9; sufficiency is verified $n \leq 7$ but not proved); and the ≤ 2 -root slack absorption. With these the Core Lemma follows modulo the Structural Lemma.